Exercise Sheet 27

All modules are finite-dimensional over $\mathbb{C}$. "Reciprocity" and the "isotypic commutant formula" refer to Lecture 27. Exercises 27.1–27.3 concern invariant tensors. Exercises 27.4–27.6 concern generation, representation rings and contractions.

Exercise 27.1 (Invariants for $\mathfrak{sl}_2$). Let $V_1 = \mathbb{C}^2$ with basis e_1, e_2, and let $\beta(u,v) = \det(u\ v)$.

(i) Decompose $V_1^{\otimes 3}$ and $V_1^{\otimes 4}$. Find $\dim \mathrm{End}_{\mathfrak{sl}_2}(V_1^{\otimes 2})$ and $\dim \mathrm{End}_{\mathfrak{sl}_2}(V_1^{\otimes 3})$.

(ii) Deduce $\dim (V_1^{\otimes 4})^{\mathfrak{sl}_2}$ and $\dim (V_1^{\otimes 6})^{\mathfrak{sl}_2}$ using duality and the identification of intertwiners with invariants.

(iii) Let $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1$ and $E(u \otimes v) = \beta(u,v)\Omega$. Let $P(u \otimes v) = v \otimes u$. Prove $1 - P = E$ on $V_1 \otimes V_1$. Why does this not contradict the three matchings of four points?

Exercise 27.2 (Degree-three invariants for $\mathfrak{sl}_3$). Let $V = \mathbb{C}^3$ and $\mathfrak{g} = \mathfrak{sl}_3$.

(i) Prove $\dim (V^{\otimes 3})^{\mathfrak{g}} = 1$, and show that vol spans it. Is vol invariant under $\mathfrak{gl}_3$?

(ii) Prove $(V \otimes V \otimes V^*)^{\mathfrak{g}} = 0$.

(iii) Prove that the invariant trilinear forms on $\mathfrak{g}$ form a two-dimensional space with basis

$$d(Y,Z,W) = \operatorname{tr}\big(Y(ZW + WZ)\big), \qquad f(Y,Z,W) = \operatorname{tr}\big(Y[Z,W]\big).$$

Exercise 27.3 (Volume tensors).

(i) Show that $g^{\otimes N}\mathrm{vol} = \det(g)\,\mathrm{vol}$ for $g \in GL_N$, and that $X \in \mathfrak{gl}_N$ acts on vol by $\operatorname{tr} X$.

(ii) For $\mathfrak{so}_3$ on $V = \mathbb{C}^3$, prove $(V^{\otimes 3})^{\mathfrak{so}_3} = \mathbb{C}\,\mathrm{vol}$ and $(V^{\otimes 3})^{O_3} = 0$. Identify the matching intertwiner $V \otimes V \to V$.

(iii) Let $\omega = \sum_i e_i \wedge f_i \in \Lambda^2 \mathbb{C}^{2n}$. Prove $\omega^{\wedge n} = n!\, e_1 \wedge f_1 \wedge \cdots \wedge e_n \wedge f_n$.

(iv) For $N = 4$, prove $\dim \mathrm{End}_{\mathfrak{so}_4}(V \otimes V) = 4$. Prove that the elements that also commute with $r^{\otimes 2}$, where $r = \mathrm{diag}(-1,1,1,1)$, are the span of $1, P, E$.

Exercise 27.4 (Faithful but not tensor-generating). A module V *tensor-generates* if every simple module is a summand of some $V^{\otimes k} \otimes V^{*\otimes l}$.

(i) Prove that every weight of every such summand lies in the subgroup $\Lambda_V \subseteq P$ generated by the weights of V.

(ii) Show that the adjoint $\mathfrak{sl}_2$-module V_2 is faithful but does not tensor-generate. In $K(\mathfrak{sl}_2) = \mathbb{Z}[x]$, show that the subring generated by $[V_2]$ does not contain $[V_1]$.

(iii) Show the same for the vector module of $\mathfrak{so}_N$, $N \geq 5$.

Exercise 27.5 (The representation ring of $\mathfrak{sl}_3$). Let $V = \mathbb{C}^3$ be the defining module. Let $x = [L(\omega_1)] = [V]$ and $y = [L(\omega_2)]$ in $K(\mathfrak{sl}_3)$.

(i) Express $[L(2\omega_1)]$, $[L(\omega_1 + \omega_2)]$ and $[L(3\omega_1)]$ as polynomials in x, y. Check the dimensions.

(ii) Decompose x^3, and compare it with Schur–Weyl duality. What is $\dim \mathrm{End}_{\mathfrak{sl}_3}(V^{\otimes 3})$?

Exercise 27.6 (Orthogonal and symplectic contractions). Use the operators $1, P, E$ on $V \otimes V$. In the symplectic case set $E(u \otimes v) = \beta(u, v)\Omega_{\mathrm{sp}}$; in the orthogonal case use the definition in Lecture 27.

(i) For $\mathfrak{sp}_{2n}$, prove $E^2 = 2nE$ and $PE = EP = -E$. Verify that $p_0 = \frac{1}{2n}E$, $p_S = \frac{1}{2}(1 + P)$ and $p_A = \frac{1}{2}(1 - P) - \frac{1}{2n}E$ are orthogonal idempotents with sum 1.

(ii) Compute their ranks from traces. Check the case of $\mathfrak{sp}_4$.

(iii) Repeat the trace computation for $\mathfrak{so}_N$. Explain the case $n = 1$ in (ii).